Hardware User Manual

MLII Series

PCI-R1604-MLII

AJINEXTEK AXT

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Revision History

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	This symbol indicates Caution (includes Warning) regarding the information. Any disregard of this indication may lead to board damage and/or defect leading malfunction.

	This indicates reference and/or information regarding the use of board.

1. Contents

1.1. Introduction

This product is a Mechatrolink-Ⅱ (MLⅡ) communication protocol support PCI half size board with control for standard 16 axis/maximum 30 axis. PCI-R1604-MLII Board uses the MLII Communication Protol to communicate in realtime and distributed control. This product can be used for precision control of main actuators such as step motor, DC servo motor, and AC servo motor for industrial equipments such as semiconductor manufacturing equipment, textile equipment, packaging equipment, and PCB engraver. PCI-R1604-MLII can connect up to 30 MLII Modules and mount up to 16 motion function modules. An installation of expansion module can allow the mounting of a maximum of 30 module function modules. MLII Slave Module includes the Motion Function Module (SGDV-xxxx11/15, JEPMC-PL2910), Digital Input and Output Module (JEPMC-IO2310), and Analog Input and Output Module (JEPMC-AN2910, JEPMC-AN2900).

1.2. Functions and Characteristics

▶ MLII Connection Form

30 Function Modules installable in two different ports.

Up to 16 motion function modules are installable in standard (Installation of option board allows up to 30)

▶ Motion Synchronization Drive

Selective Axis synchronized drive for up to 30 axes through program. 2-4 axes linear/2 axes circular interpolation possible. 3 axes helical interpolation.

▶ Motion Drive Method

Fixed distance drive, jog drive, signal detection drive, interpolation drive, MPG drive

Speed and Distance override function in operation

▶ Motion Drive Adjustable Speed Mode

Symmetrical trapezoid, symmetrical S-curve, Asymmetrical trapezoid, Asymmetrical S-curve drive support

▶ Advanced Motion Drive

Servo Driver Control Speed Drive/Torgue Drive Funcntion, Realtime Position Compensation function, ECAM function

▶ Communication Standard

Communication Cycle: 1 mSec, 500 uSec, 250 uSec Selective, 17-byte mode,

Communication Ports: 2 EA, Maximum number of nodes for port: 15 EA (1 mSec if communication setting is made)

Motion Control Function and Communication Control Cycle are planned irrelevant to the number of control axes.

▶ Firmware Upgrade Function

Update Function internally installed for user's request of function generating.

1.3. Specifications

Diagram 1. PCI-R1604-MLII Mechatrolink-II master board specification

Item	Specification
Support Communication Protocol	Mechatrolink-II (YASKAWA Standard of Distributed Networ
MLII port	2 port
MLII Communication Specificaications	17-byte, 10 Mbps fixed, 1mSec, 500uSec, 250uSec selective communication cycle
Motion Generation Chipset	CAMC-QI 4EA Standard Mounting
Connection	USB type A, Exclusive STP cable(C.I. : 130Ω) Minimum: 0.5m(per node), Maximum: 50m(Total)
Motion Instruction Speed	~10.0000Mpps Command Posiotion Change Rate
Motion Position Range	32-bit Range Counter(same command/actual)
Maximum Number of Installable Axes	Standard 16 Axes, 30 Axes for Option Board Installed
LED Display	Comm (green), Error (red)
BUS Connection Method	PCI 2.2, Memory access, 32 bits
Operating Temperature Ran	0 ~ +50℃
Operating Humidity Range	80% Below
Dimension	174.63 × 106 mm
Front Panel Connector	USB Connector (NET#1, NET#2, 1 per each)
Synchronization Begin/Stop between Boards	J3/J4 3-pin 2.0mm

Operating Power:	
5V(Allowable Current: 4.7V t 5.3V)	1.5A(Maximum Current Consumption: 1.6A)

2. Board Specification

2.1. Hardware BLOCK DIAGRAM (Block Diagram)

PCI-R1604-MLII is larged dived into 6 blocks: 1) CPU (DSP) that performs for motion control and communication manange of Master Board, 2) Chipset for MLII Communication (YASKAWA ASIC Chip for MLII), 3) Trasceiver that transmits communication signal, 4) DPRAM that saves and loads data, 5) FPGA that transfer data between PCI and DSP, and PCI and CAMC-QI, and 6) CAMC-QI that performs for motion control.

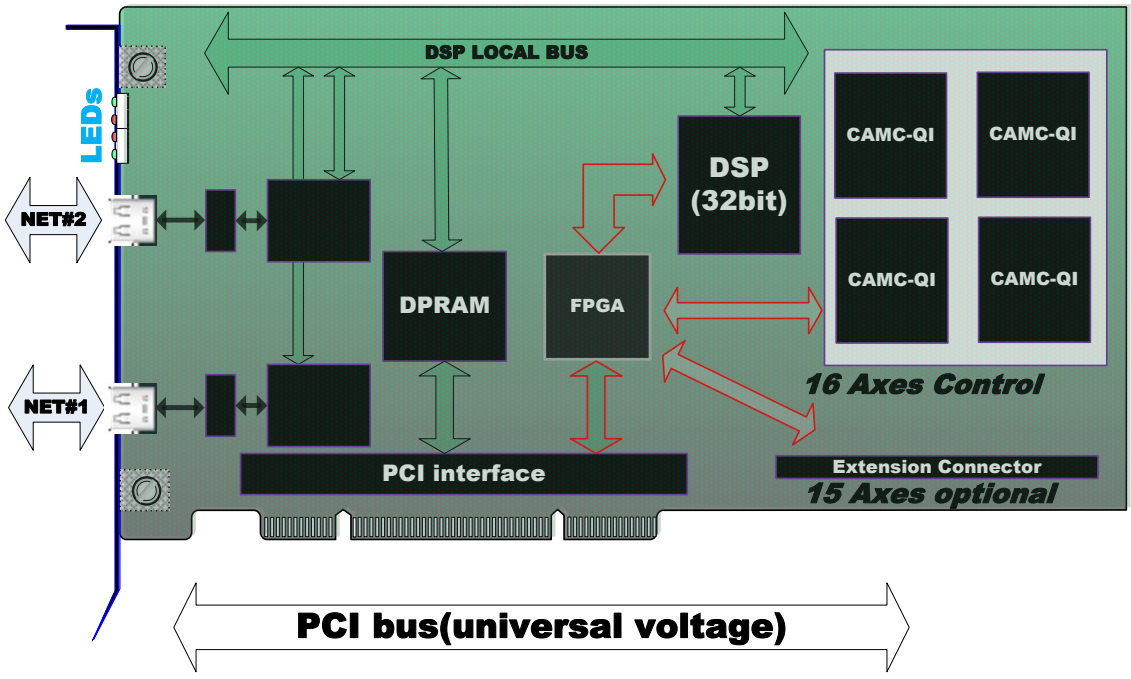


Diagram 1. PCI-R1604-MLII Block Diagram

PCI-R1604-MLII consistent maintains motion control and communication control performance levels despite the number of axes and PC system performance through the use of independent H/W (CAMC-QI) for motion control.

2.2. Hardware Information

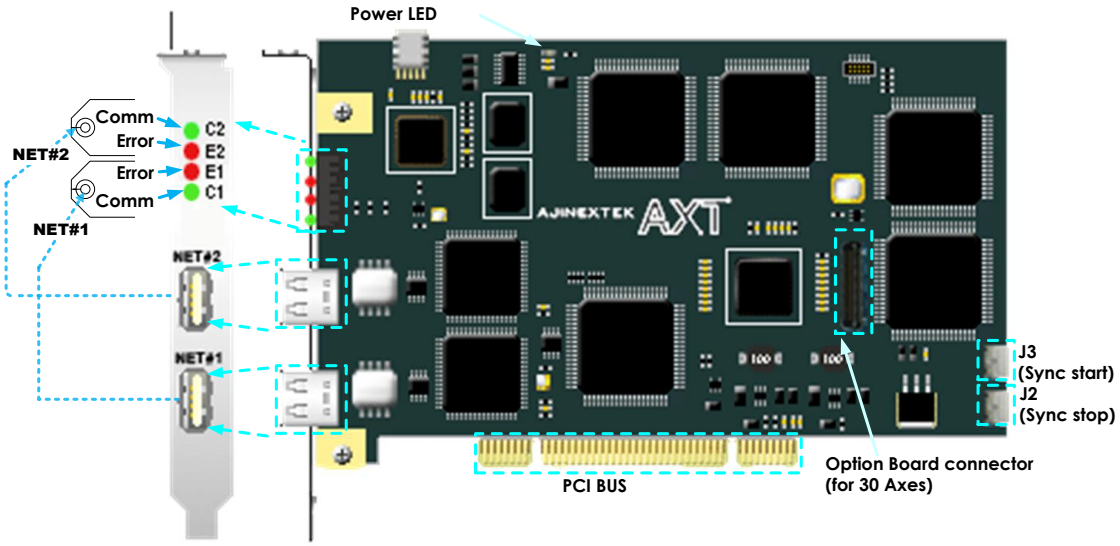
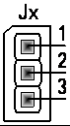
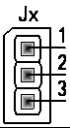


Diagram 2. PCI-R1604-MLII master board and connector

Diagram 3 shows the external connector and indication LED of PCI-R1604-MLII.

The board consists of Status LED (Comm, Error) for the operation status of board, Synchronization Signal Connetion Connector for motion synchronization between boards, USB Connector (NET#1, NET#2) for the MLII Network Connection, and Connector to the Option Board for expansion of standard 16 axes motion control into 30-axes control capable board.

Signal	Function	Information
C1,E1 C2,E2	C1(Comm), Green	LED indicates current Connection Status of ET#1.
	E1(Error), Red	LED indicates current connection error status ET#1의
	C2(Comm), Green	LED indicates current connection status of ET#2.
	E2(Error), Red	LED indicates current connection error status ET#2.
J3(2mm pitch) 	Sync start	1 : Sync start group 1 Input/Output 1 : Sync start group 2 Input/Output 3 : GND
J2(2mm pitch) 	Sync stop	1 : Sync. stop group 1 Input/Output 1 : Sync. stop group 2 Input/Output 3 : GND
NET#1	MLII port #1	Port #1 of MLII Communication (15 Slaves

		onnectable)
NET#2	MLII port #2	Port #2 of MLII Communication (15 Slaves onnectable)
J4	Optional board connector	Connect to the Optional Board (PCI-E1604) fo 0-axes control

LED indication method for status is shown below.

Status	Cx (cycle)	Ex (cycle)
Communication H/W Error Lv1.	On ➔ Off Repeat (2sec)	On ➔ Off Repeat (2sec)
Communication H/W Error Lv2.	On ➔ Off Repeat (4sec)	On ➔ Off Repeat (4sec)
Slave not connected	On ➔ Off Repeat (2sec)	On
Slave connection, no error, normal state	**On ➔ Off Repeat (0.5sec)**	**Off**
Communication H/W Error after Slave is Connected	On ➔ Off Repeat (0.5sec)	On ➔ Off Repeat (0.5sec)
Communication Error with Slave after Slave is Connected	On ➔ Off Repeat (0.5sec)	On ➔ Off Repeat (1sec)
Real Scan and Lock information does not match with Configuration Lock status	On ➔ Off Repeat (0.5sec)	On ➔ Off Repeat (1sec)

Caution) Refer to "4.4 Resolving Connection Errors" for all errors

Caution) The LED blinking cycle is applied in case of the communication cycle is set at 1 mSec. The LED blinking cycle is applied in half the cycle time in case of communication cycle is set at either 500 uSec or 250 uSec.

3. Installation

3.1.Hardware Installation

The consisting parts of product must be installed in the following order of instructions. The product consists of PCI-R1604-MLII board, MLII Slave Module (SGDV-xxxx11/15, JEPMC-PL2910, JEPMC-IO2310, JEPMC-AN2910, JEPMC-AN2900), and Cable for MLII.

Install the PCI-R1604-MLII into the PC in the following order.

① Ensure to condence all electric charges before installing the PCI-R1604-MLII.

② Turn off the power switch of IPC and Pull the power plug.

③ Select the PCI slot for installation.

④ Remove the panel of the empty slot in order to install the PCI.

⑤ Insert the board into the selected PCI slot. Fix the bracket onto the IPC using a screw.

⑥ Ensure to check for proper installation and make certain the board has been inserted firmly into the PCI connector.

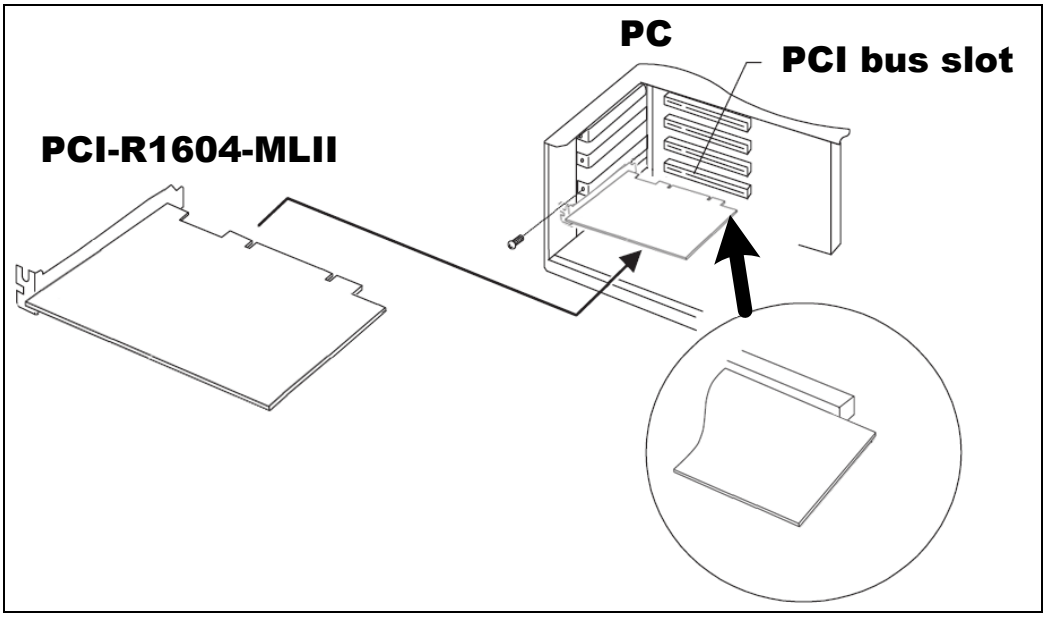


Diagram 3. PCI Slot Installation

⑦ Plug in the IPC power plug and turn on the power switch in order to operate the system.

⑧ Ensure to check the the C1/C2 LED are blinking properly located external Bracket.

⑨ Ensure to check the system is booting properly.

3.2. Software Installation

Upon proper installation of the board, install the software using the following set of instructions.

① Prepare the ExSoftware program from either the CD provided with the product or downloaded from the website. Recent version of software is available on the website. (Website: http://www.ajinextek.com/, Support: Refer to Download)

② Run the "Setup.exe" from the prepared ExSoftware Installation Program.

③ Once the installation is complete, install the device driver in the following set of instruction.

- Check for the board shown below in the Device Manager.

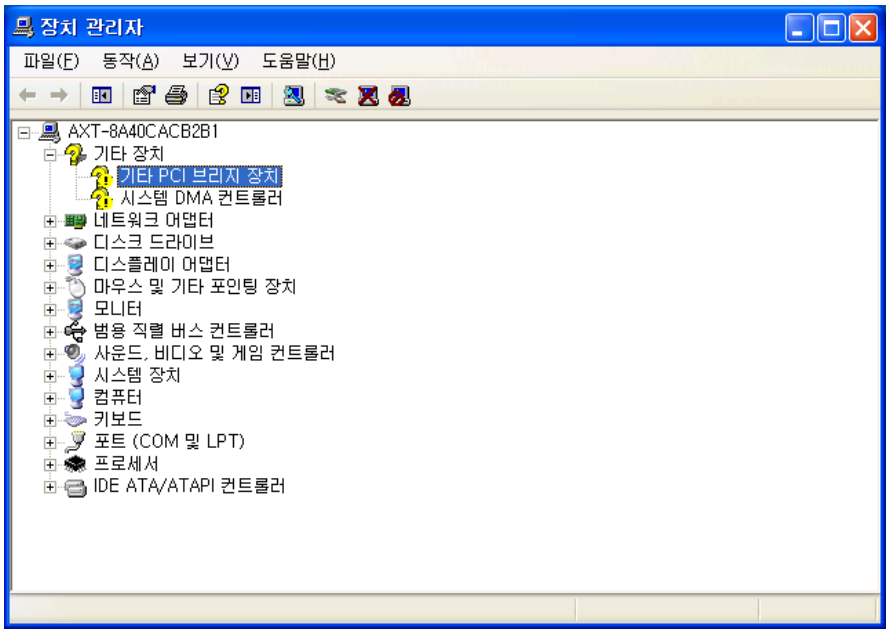


Diagram 4. Device Manager in Initial Recognition

Normally, PCI-R1604-MRII is recognized as "Other Device/Other PCI Bridge Device"

- Click right button on the mouse and select “Driver Update” from the menu.

-

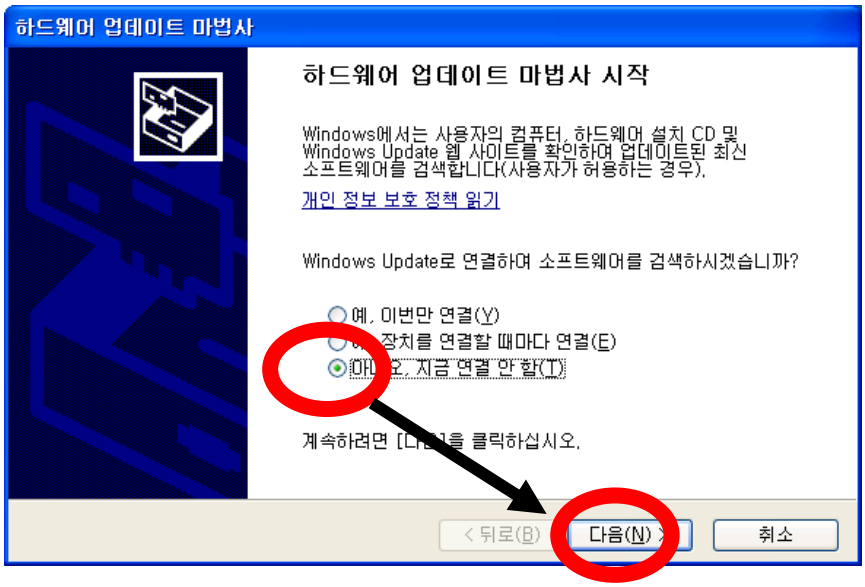


Diagram 5. Hardware Update Location Setting #1

- Next, install the software.

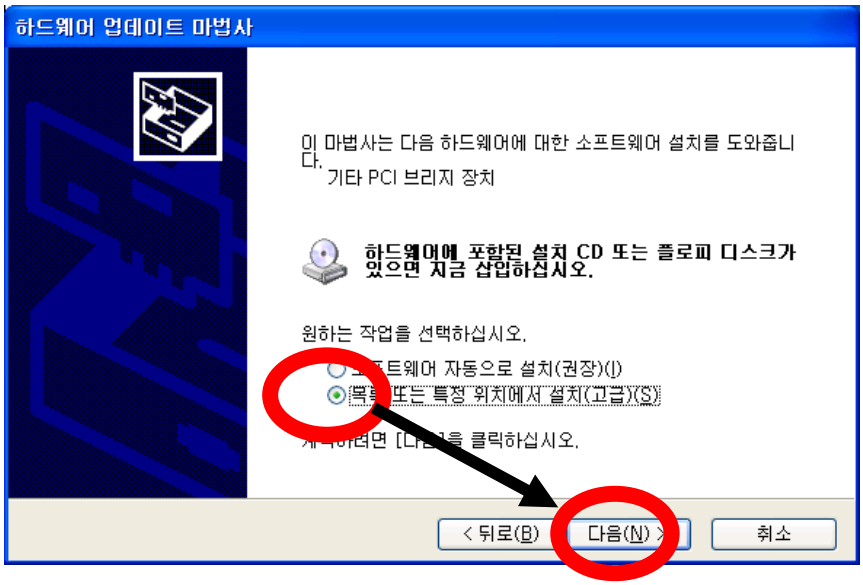


Diagram 6. Hardware Update Location Setting #2

- Next, select the search location for the device driver. Insert the following location in search with "C:₩Program Files₩EzSoftware₩INF(Driver)" and look inside the "INF(Driver)" for the driver.

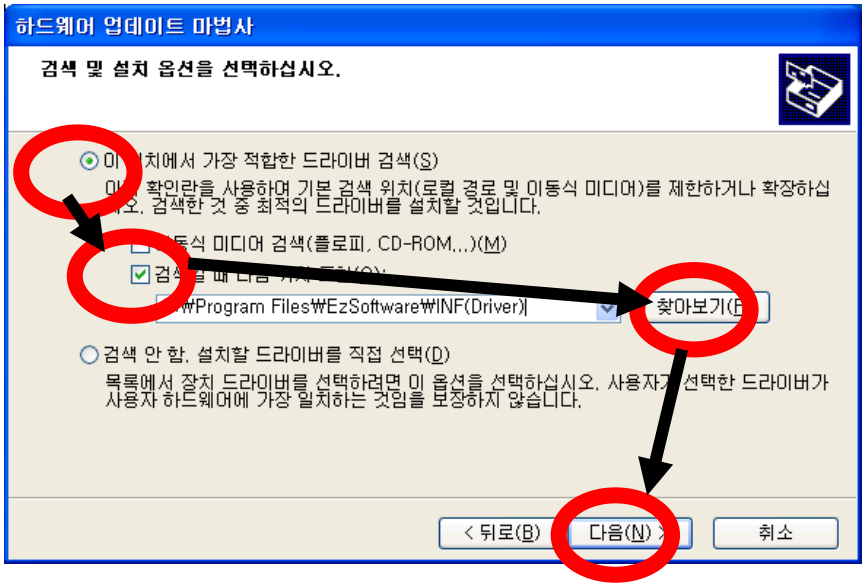


Diagram 6. Hardware Update Location Setting #3

Proceed when the EzSoftware is properly installed.

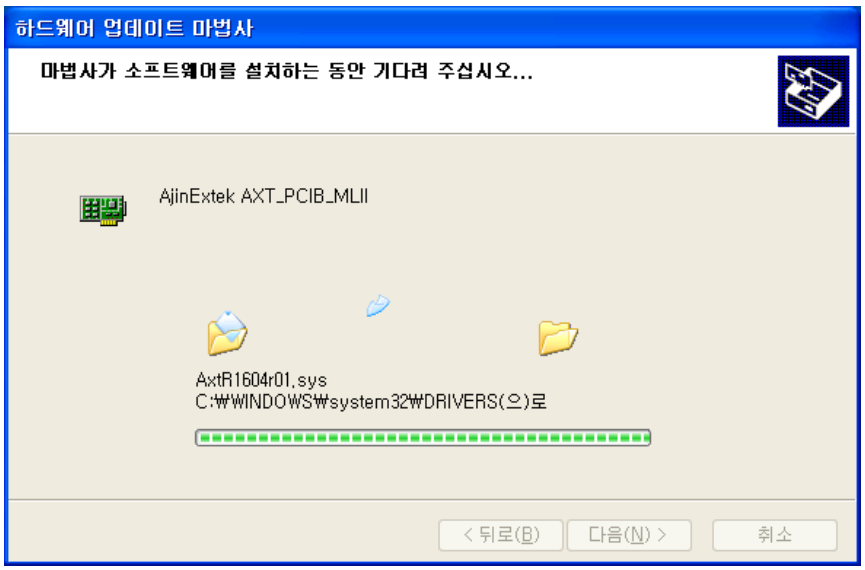


Diagram 7. Needed Files Copy Process

- The window will display the following if the device driver has been installed properly.

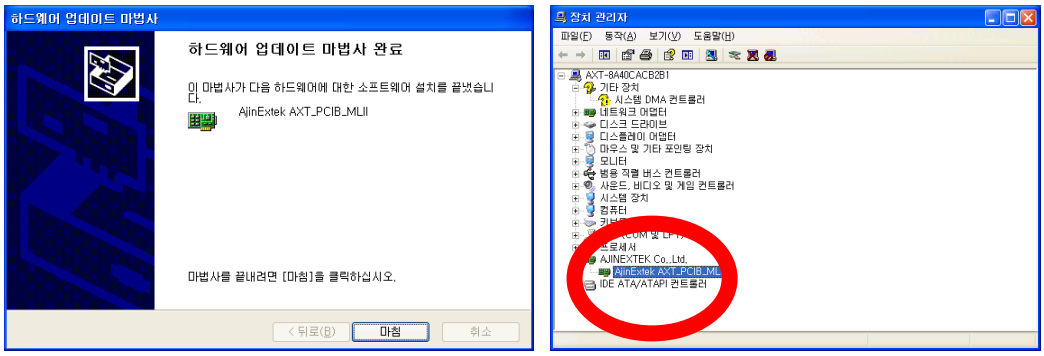


Diagram 8. Hardware Update Complete

④ It is necessary to repeat the installation procedure if there are several PCI-R1604-MLII installed for each board.

⑤ If the device driver has been properly installed in the Device Manager, run the EzConfig program and check on the status of PCI-R1604-MLII.

Caution) If the Slave node has been connected after the PCI-R1604-MLII has been supplied power, the Scan in EzConfig must be made before checking for the connection status.

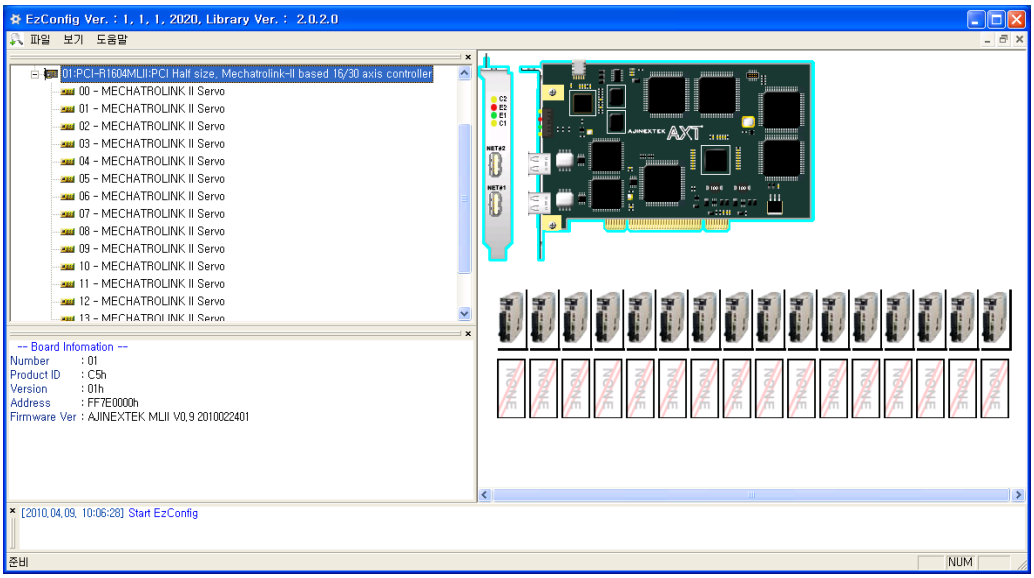


Diagram 9. Screen of Normal Hardware and Software Installation
(The display may be different for the Slave connection status)

4. BOARD CONNECTION

4.1. External Connection

PCI-R1604-MLII is connecting with the Slave modules in Daisy-chain form using the exclusive cable in the following diagram.

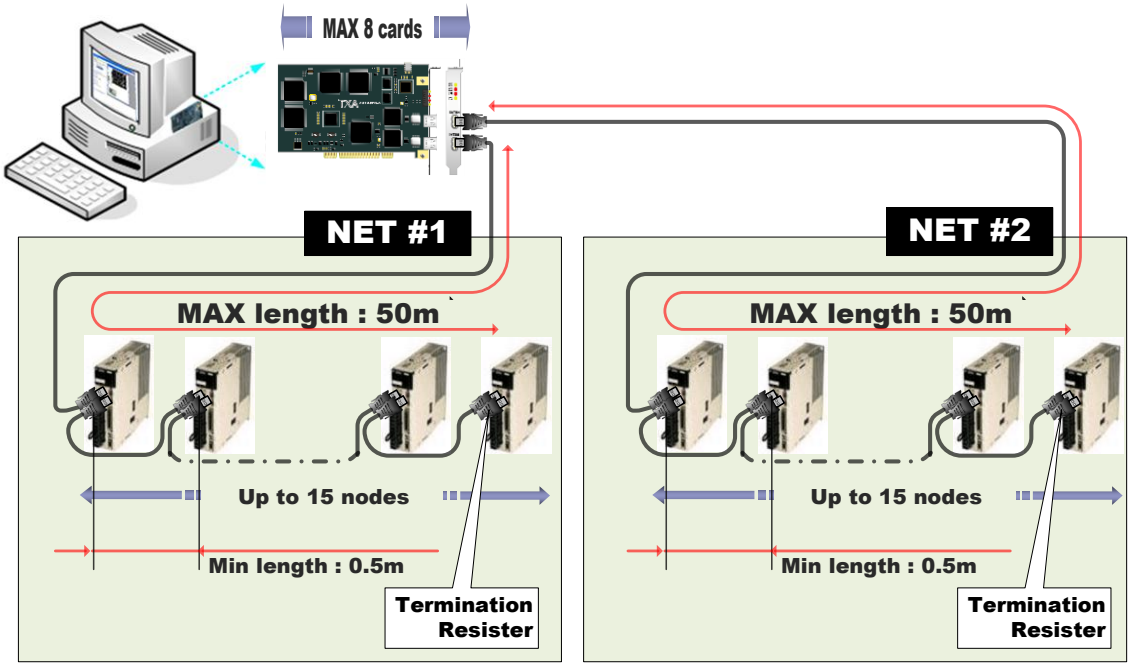


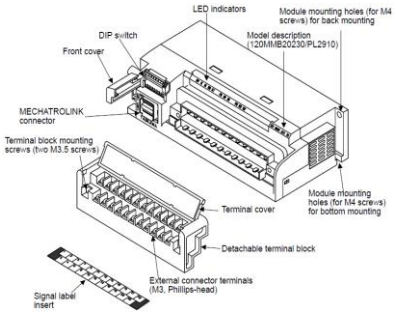
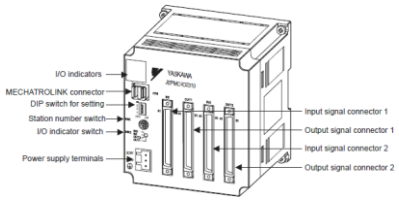
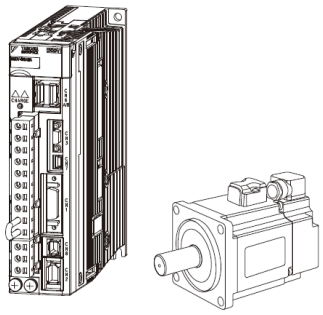
Diagram 10. MLII Network Connection

PCI-R1604-MLII allows an installation of up to 30 Salve Modules for a maximum of 15 per NET, and a standard of 16 for the dual-motion function module (SGDV-xxxx11A). For additional installation for more than 16 axes of motion control, PCI-E1604 must be additionally installed at J4. The Slave of each NET can be configured as independent and does not always require use. The Slaves connected to NET#1 will be assigned lower Axis Number and Terminal Number than those of NET#2. Therefore, in case of installing two Servo drivers to NET#1 and 5 servo drivers to NET#2, the axis number for the servo of NET#1 will be assigned 0,1 and NET#2 will be 2,3,4. The Station Address in the same NET will be assigned in ascending order accordingly to the axis and terminal numbers.

Last Slave at the end of each NET must be installed with Termination Resistor. Nodes must have a minimum of 0.5M in distance and each NET's maximum connection distance is below 50m. All Slaves must be set at 17-byte, 10Mbps communication mode, and setting method is explained in detail in the MLII Slave Manual.

4.2. Slave Modules connecting to PCI-R1604-MLII

The following table shows the products connectable to PCI-R1604-MLII. Any other product requested for listing may be added.

(Maker) Product (Producer)	Function	Form
JEPMC-PL2910 (YASKAWA)	- Pulse Output Module - Max. 500kHz - Control Axes : 2축 - No Encoder Input - Limited Simple P to P - Additional 5V Power Supply Needed - Power : 24V	
JEPMC-AN2900 (YASKAWA)	- Analog Input - Channels: 4CH - ±10V, 16-bit resolution. - 1mSec Sampling - Input Delay Time: Max. 4 mSec - Power : 24V	
JEPMC-AN2910 (YASKAWA)	- Analog Output - Channels: 2CH - ±10V, 16-bit resolution. - Max. Loading Current: ±5mA (2 KΩ) - Input Delay Time: Max. 4 mSec - Power: 24V	
JEPMC-IO2330 (YASKAWA)	- Digitial Input and Output - Terminals : 64 Inputs, 64 Outputs - Output Drive Current : 50mA/Terminals - Power: 24V	
SGDV-XXXX11A (YASKAWA)	- Servo Driver - Supported Motors: SGMJV, SGMAV, SGMPS, SGMGV, SGMSV, SGMCS - Power : AC100,200,400	

Note) Refer to the Slave Module Manual for each Slave Module

4.3. CABLE / Connector

The communication cable connecting the PCI-R1604-MLII and Slave module must be exclusive STP Cable. The minimum connection distance between the nodes is 0.5m, maximum cable distance per Network is 60m. Connector may look as a simple USB cable, it has a Lock method, and hence an exclusive cable must be installed.

⚠	Mechatrolink-II communication method for motor control is a method in which high-speed data is transmitted and received through communication line. Motor control is made with the transmitted data; therefore high reliability in the communication line is required. It is important to use the connection cable for this purpose. Using a simple, abnormal cable may result in communication error, leading to motor runaway, control failure, equipment damage and user injury.

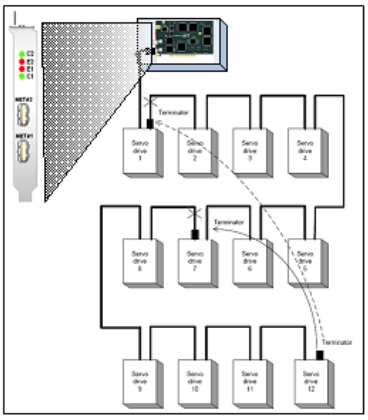


Diagram 11. Cable and Connector Specification for MLII

(Caution) Regular USB cable is not operational with this equipment.

4.4. Resolving Connection Error

E1/E2 LED blinks in case of communication connection error. In order to find the error in communication line, remove the terminating resistor and place it in each node until the Error LED turns off. Once the Error LED turns off, the node can be the source of error in communication. Upon this finding, examine the communication cable, connector, and connection status of the nodes.



Caution) In order to check on the changes to node status

in the communication line, the concerned board must be set at "Configuration Unlock" in EzConfig, and perform Scan in the File Menu upon the Connection change .

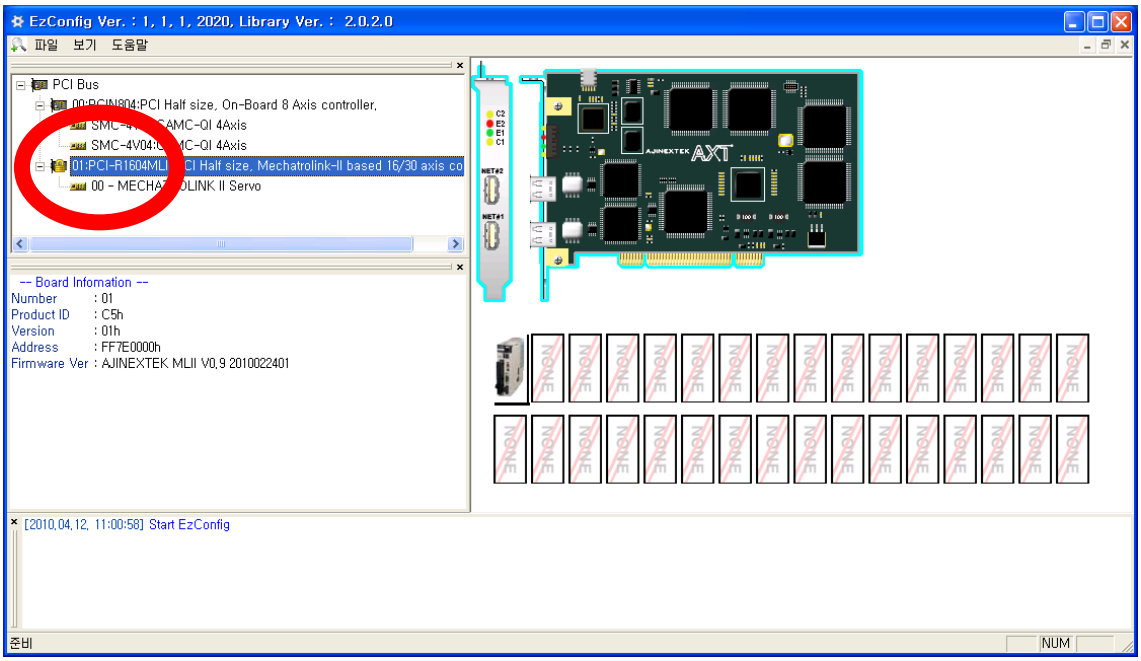


Diagram 12. Display of EzConfig in Configuration Lock (Check on the Padlock Icon) (Network Initialization in Lock State uses saved information instead of real-time current information.)

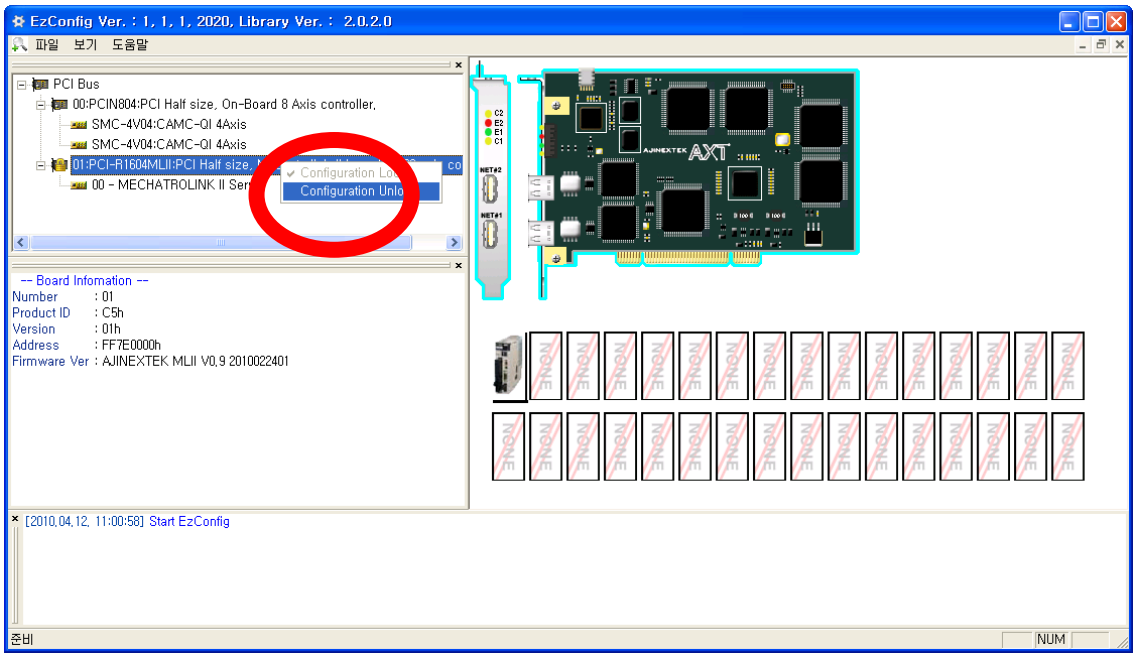


Diagram 13. Configuration Unlock Method (Click on the right button of mouse).

(Unlock must be made for the Slave Node Connection Status changes to be reflected.)

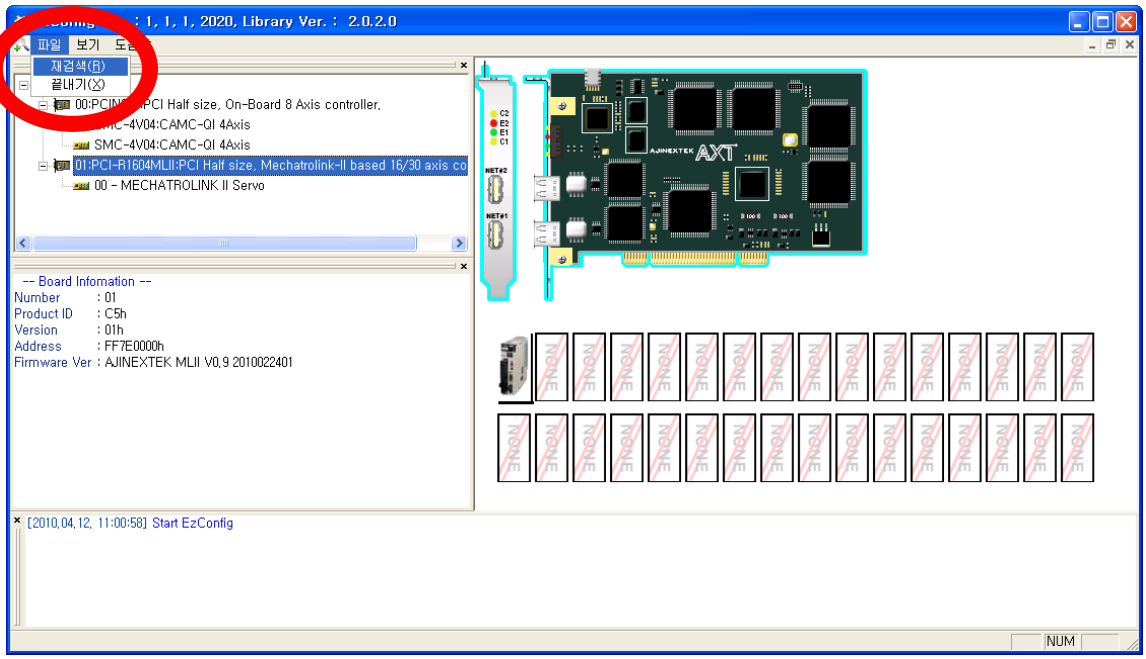


Diagram 14. Network Connection Status Scan

(Scan must be performed before updating and displaying the Slave Node changes.)

The display of PCI-R1604-MLII Connection Status in the external LED is listed below.

No.	Status	Cx (Time Cycle)	Ex (Time Cycle)
1	Communication H/W Error Lv.1	On ➔ Off Repeat (2sec)	On ➔ Off Repeat (2sec)
2	Communication H/W Error Lv.2	On ➔ Off Repeat (4sec)	On ➔ Off Repeat (4sec)
3	Slave is not connected	On ➔ Off Repeat (2sec)	On
4	Slave is connect and normal status	On ➔ Off Repeat (0.5sec)	Off
5	Communication H/W error after Slave is connected	On ➔ Off Repeat (0.5sec)	On ➔ Off Repeat (0.5sec)
6	Slave Communication error after slave is connected	On ➔ Off Repeat (0.5sec)	On ➔ Off Repeat (1sec)
7	Lock information does not match the real time current scan information in Configuration Lock	On ➔ Off Repeat (0.5sec)	On ➔ Off Repeat (1sec)

Caution) LED blinking cycles listd above only apply to the communication cycle that is set at 1 mSec. The LED blinking cycles are halved from the above listed cycles for Time Cycles set at either 500 uSec or 250 uSec.

Resolutions are listed in the following.

No.	Status	Resolutions
1	Communication H/W Error Lv. 1	-. Check for firm installation into PCI Slot. -. Check on board power LED. -. Request After-service.
2	Communication H/W Error Lv. 2	-. Check for firm installation into PCI Slot. -. Check on board power LED. -. Request for after-service.
3	Slave is not connected	-. Check the connection with Slave. -. Check for the Communication Mode Setting of Slave. -. Check if Scan in the File Menu of EzConfig has been performed
4	Slave is connected and normal status	-
5	Communication H/W error after Slave is connected	-. Check the Communiction line with Slave. -. Check on the Terminating Resistor in the last Slave. -. Check the Slave Communication Setting.(Station Address and Byte mode) -. Check on the power to Slave. -. Request for after-service.
6	Slave Communication error after slave is connected	-. Check the Communiction line with Slave. -. Check on the Terminating Resistor in the last Slave. -. Check the Slave Communication Setting.(Station Address and Byte mode) -. Check on the power to Slave.
7	Lock information does not match the real time current scan information in Configuration Lock	-. Check if the real-time current status and EzConfig Lock information match. < Changing the Lock Information > > Configuration Unlock ➔ Scan ➔ Configuration Lock
8	Other errors	-. Check for firm installation into PCI Slot. -. Check on board power LED. -. Request After-service.

Caution) PCI-R1604-MLII automatically performs initialization for the connection status on Slave once after initial power on. If the Configuration Lock is set for the Slave setting, the MLII Network Initialization will be performed with the previously saved information. Therefore, Slave Node connection changes after PCI-R1604-MLII power on require the Configuration Unlock status by Scan in the File Menu of EzConfig in order for current Slave connections to be displayed.



Reference) Configuration Lock/Unlock Function

Disrtibuted Control Component through the Network may result in errors in the communication status due to problems with the communication line and/or slave. In errors concerning the Slaves, Axes and Terminal numbers that the user assigned may be re-assigned and operate motor and output terminals that are unintended, resultings in serious issues. As a preventive measure, it is necessary for the information regarding the Network Control Components be saved by the user and the location of control factor used by the user to be fixed.

In order to resolve the above mentioned issue, "Configuration Lock" function may be used in the EzConfig to save the control component information into the nonvolatile memory. Once the "Configuration Lock" function applied, the saved information regarding the fixed factors (axes and terminal numbers) are automatically managed despite the network Communication error and/or Slave errors.

4.5. Communication Cycle Setting

The Communication Cycle of PCI-R1604-MLII may be set in the following and each communication cycle limits the number of mountable slaves. Setting faster communication cycle speeds up the data transmission for slave, but it decreases the controllable number of slave mountable.

No.	Communication Cycle	Max. Slaves Mountable for each Net	Max. Number of Slaves Mountable
1	1 mSec(default)	15	30
2	0.5 mSec(500 uSec)	7	14
3	0.25 mSec(250 uSec)	2	4

Caution) Scan in the File Menu must be performed after the Communication Cycle has been modified. Even if the Slave setting has been set at "Configuration Lock", setting the Communication Cycle automatically sets "Configuration Unlock," and "Configuration Lock" will have to be set again.

Caution) Communication Cycle Setting of PCI-R1604-MLII does not require any special setting to the Slave.

The Communication Cycle can be set as shown in the diagram below. The setting information will be automatically saved even if the power has been shut. The previously set communication cycle will be operational after the power has been supplied again.

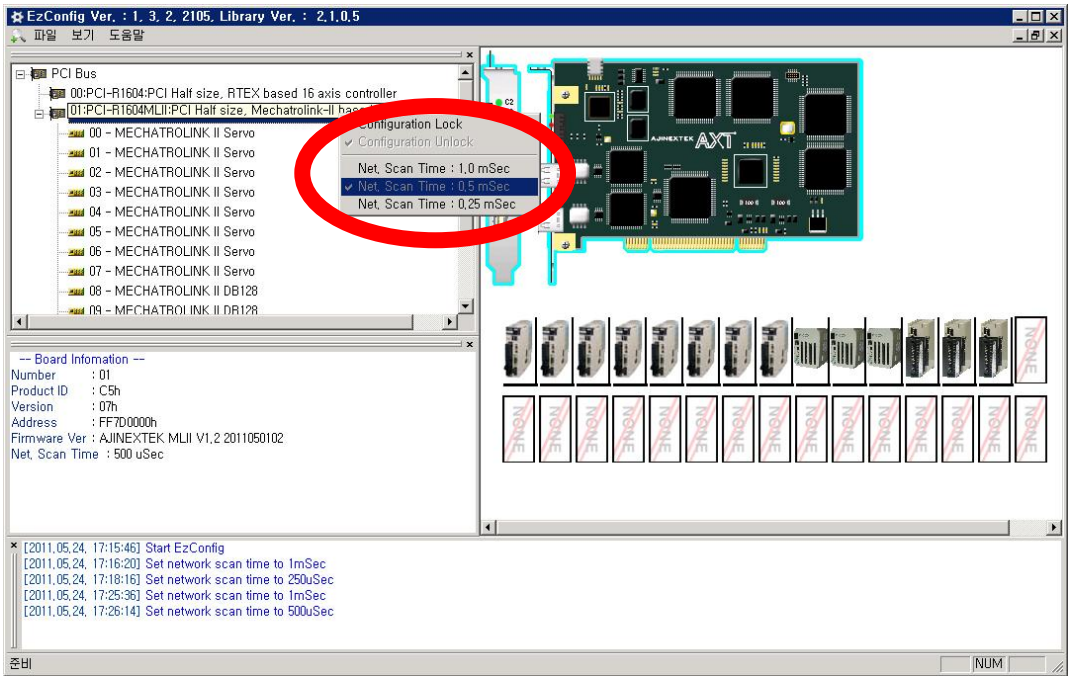


Diagram 15. Setting the Communication Cycle

(Scan must be performed after setting the Communication Cycle)

5. Ordering Information for Related Products

5.1. Master card

Product	BUS Method	Number of Installable Nodes (Number of Control Axes)	Applied System and/or Communication
PCI-R1604-MLII	PCI Half Size	30(16 axes)	Industrial PC, regular PC, Mechatrolink-II
PCI-E1604	Local bus	(16 axes)	Optional Board installable to PCI-R1604-xxx

5.2. Communication Cables / Teminator

Product	Length(m)	Other
JEPMC-W6002-A5	0.5	Mechatrolink-II USB connector to USB connector
JEPMC-W6002-01	1	
JEPMC-W6002-03	3	
JEPMC-W6002-05	4	
JEPMC-W6002-10	10	
JEPMC-W6002-20	20	
JEPMC-W6002-30	30	
JEPMC-W6022	-	Terminator(termination resistance)

6. Terms

AnyMotion :	
	General name for Ajinextek's Motion Control Modules with control functions for all types of step motor and/or servo motor
AnyDIO :	
	General name for Ajinextek's Digital Input and Output Control Modules for all types of sensor connection functions.
AnyAIO :	
	General name for Ajinextek's Analog Input and Output Control Modules for all types of sensor connection functions.
AnyCOM :	
	General name for Ajinextek's Communication modules for all types of communication functions.
AXL :	
	Ajinextek General Library
MLⅡ(Mechatrolink-Ⅱ)	
	FieldBux Distributed Motion Control Solution provided by YASKAWA Electronics
EzSoftware :	
	Software Package for PC Installation Control Board
EzConfig, EzMotion, EzDIO, EzAI, EzAO, EzCOM Agent :	
	Software for configuration, operation and support included in ExSoftware for the functiosn of AnyMotion, AnyDIO, AnyAIO, and AnyCOM installed on the board.

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